

MA 312: Design and Analysis of Algorithms

Course Code: MA 312

Course Name: Design and Analysis of Algorithms

L-T-P-C: 3-0-2-4

Students intended for: B. Tech Maths and Computing

Elective or Core: Core

Prerequisites: IC253

Mutual Exclusion: CS212, CS403

Approval: 59th BoA

Course Contents

- **Unit I:** The Role of Algorithms in Computing, Analyzing Algorithms, Best, Average, Worst Case Analysis, Asymptotic notations, big-oh, theta, big-omega, little-oh and little-omega, Recurrences, Master's Theorem, Amortized analysis: aggregate analysis, accounting, potential method. [7 Lectures]
- **Divide and Conquer:** Merge Sort, Quick Sort, Finding Closest Pair of Points, Strassen's Algorithms, Median Finding, Discrete Fourier Transforms. [7 Lectures]
- **Greedy Algorithms:** Un-weighted Interval Scheduling, MST-- Prims and Kruskals, Dijkstra's shortest path, fractional knapsack, and Huffman coding. [7 Lectures]
- **Dynamic Programming:** Integral knapsack, Weighted Interval Scheduling, longest common subsequence, Rod-Cutting, matrix chain multiplication, and independent sets in trees. [7 Lectures]
- **Geometric Algorithms:** Line Segments: Properties, Intersection, sweeping lines- Convex Hull finding algorithms. [4 Lectures]
- **Intractability:** NP-completeness: reduction amongst problems, classes NP, P, NP-complete, and polynomial time reductions. [4 Lectures]
- **Introduction to Advanced techniques:** Approximation Algorithms, Vertex Cover, Set Cover, Minimum Makespan, Subset Sum Problem. Parameterized Algorithms, Vertex Cover, Feedback Vertex Set in Tournaments. [6 Lectures]

Laboratory/practical/tutorial Modules:

Coding Lab will cover topics discussed in this course.

Text books:

1. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, **Introduction to Algorithms**, MIT Press, 3rd Edition, 2009.
2. J. Kleinberg and E. Tardos, **Algorithm Design**, Pearson, 2006.



References:

1. Dasgupta, Papadimitriou and Vazirani, **Algorithms**, McGraw-Hill Education, 2006.
2. Horowitz, Sahni, and Rajasekaran, **Computer Algorithms**, Silicon Press, 2007.